Please ensure this tracking sheet is completed as a group, as it will be used to monitor your individual participation within the team.

Team Members	Role/ Responsibilty	Email
Ihimbazwe Samuel	DQN training algorithm (replay buffer, epsilon-greedy, Bellman target, loss, target network), play.py, Kaggle notebook, epsilon experiments	s.ihimbazwe@alustudent.com
Nkusi Orrie Dan	Environment and CNN architecture, learning rate / gamma / batch size experiments, evaluate.py, README	d.nkusi@alustudent.com

Task Allocation and Tracker

	Task Allocated	Assigned Member	Deadline	Completion Status (Not Started/In Progress/Completed/Did not participate)	Reviewed by Team? (Yes/No)	Comments
	Project setup: GitHub repo, uv environment, Pong environment and preprocessing	Ihimbazwe Samuel, Nkusi Orrie Dan	09 Nov 2026	Completed	Yes	Chose ALE/Pong-v5 together. Samuel set up the repo and uv, Dan worked on the preprocessing (84x84 grayscale, 4 stacked frames).
	train.py: CNN Q-network, online and target networks, replay buffer, epsilon-greedy with decay, Bellman target, loss, optimizer, target network updates	Ihimbazwe Samuel, Nkusi Orrie Dan	09 Nov 2026	Completed	Yes	Dan led the CNN and network setup, Samuel led the replay buffer, epsilon-greedy and the learning step. Integrated and tested together.
	play.py for live gameplay (greedy actions, real-time display)	Ihimbazwe Samuel, Nkusi Orrie Dan	10 Nov 2026	Completed	Yes	Imports the network and environment from train.py so they always match training. Reviewed by Dan.
	Kaggle GPU training notebook (uv) and baseline run	Ihimbazwe Samuel, Nkusi Orrie Dan	10 Nov 2026	Completed	Yes	Laptop CPU only did ~25 steps/s so we moved training to a Kaggle T4. Baseline showed learning starts ~200k steps, so we set 500k steps per experiment.
	10 experiments on epsilon start / end / decay, results table and analysis	Ihimbazwe Samuel, Nkusi Orrie Dan	11 Nov 2026	Completed	Yes	Best: exp09 (epsilon 0.1 to 0.01), +20 greedy. Also reran it with another seed for 1M steps to check it wasn't luck.
	10 experiments on learning rate / gamma / batch size, results table and analysis	Ihimbazwe Samuel, Nkusi Orrie Dan	12 Nov 2026	Completed	Yes	Built on Samuel's best epsilon setting. Best: exp08 (batch size 64), +20.2 greedy.
	evaluate.py and greedy scoring of all trained models	Ihimbazwe Samuel, Nkusi Orrie Dan	12 Nov 2026	Completed	Yes	Scores a model with epsilon = 0 without opening a window, used to compare all 20 runs.
	README: project intro, network architecture diagram, training process, gameplay video	Ihimbazwe Samuel, Nkusi Orrie Dan	14 Nov 2026	Completed	Yes	Dan leads the README, Samuel adds the evaluation and gameplay parts.
	Final model choice, task sheet, submission, presentation slides and booking	Ihimbazwe Samuel, Nkusi Orrie Dan	15 Nov 2026	Completed	Yes	Comparing the best models on the same games. Dan's exp08 never got stuck in a rally loop, so it is the likely demo model.

Meeting Attendance Log

Meeting Date	Agenda	Facilitator	Attendees	Key Discussion Points	Next Steps
08 Nov 2026	Kick-off: read the assignment and split the work	Ihimbazwe Samuel	Ihimbazwe Samuel, Nkusi Orrie Dan	Went through the requirements and rubric, agreed on roles, compared Atari games and leaned towards Pong because DQN learns it reliably.	Set up the repo with uv, start train.py.
09 Nov 2026	Environment choice and train.py	Ihimbazwe Samuel	Ihimbazwe Samuel, Nkusi Orrie Dan	Confirmed Pong. Went through the CNN, replay buffer and training loop and tested them together. The laptop was too slow to train (~25 steps/s), so we decided to train on Kaggle GPUs.	Run a 500k-step baseline on Kaggle.
10 Nov 2026	Baseline results and experiment plan	Ihimbazwe Samuel	Ihimbazwe Samuel, Nkusi Orrie Dan	Baseline only started learning around 200k steps, so we fixed 500k steps for every experiment. Agreed on the fixed settings and the results table format. Started the experiments.	Each member runs their 10 experiments.

11 Nov 2026	Review of the first experiment results	Ihimbazwe Samuel	Ihimbazwe Samuel, Nkusi Orrie Dan	Samuel's exp09 (epsilon 0.1 to 0.01) won +20. Noticed we had both tuned epsilon, so Dan switched to learning rate / gamma / batch size using exp09's epsilon, and the notebooks were switched to uv only.	Dan reruns his experiments, Samuel runs exp10 and a second-seed check of exp09.
12 Nov 2026	Final results, model choice and documentation	Ihimbazwe Samuel	Ihimbazwe Samuel, Nkusi Orrie Dan	Compared the best models on the same games. The 1M-step rerun got stuck in one endless rally, Dan's exp08 won every game. Planned the README, task sheet and presentation.	Finish README, slides and Q&A prep, submit on 15 Nov.